# Stochastic Modeling
# Regime-Based Rotation Strategy

A Regime-Switching Approach to Tactical Asset Allocation Using VIX and HMMs

## 1 Introduction and Scenario

Our team on the Quant Strategies Desk has been tasked with designing a regime-based allocation strategy. The objective is to rotate among three ETFs—TLT (long-term U.S. Treasuries), GLD (gold), and SPY (S&P 500)—by inferring the market's volatility regime from the VIX index.

This report details the step-by-step process of developing and evaluating this strategy, from data preparation to backtesting. The core hypothesis is that the performance of these asset classes is conditional on the volatility environment, and a model that can identify these environments can lead to superior risk-adjusted returns.

## 2 Step 1: Data Preparation and Exploration

### 2.1 Approach

We downloaded daily adjusted close prices for TLT, GLD, SPY, and the VIX index from Yahoo Finance. To ensure data consistency, we used the maximum common sample period and removed any dates with missing values. Daily log-returns were calculated for the three ETFs, and both the daily change ($\Delta$VIX) and log-return (VIX ret) were computed for the VIX.

### 2.2 Results and Observations

The final clean dataset spans from **November 22, 2004, to February 20, 2026**, comprising 5,345 trading days.

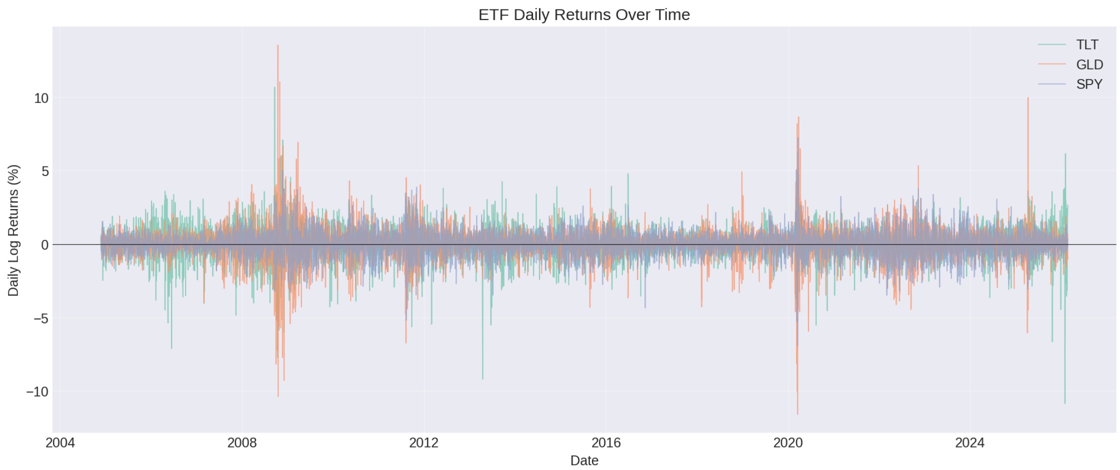


Figure 1: Daily Log-Returns of TLT, GLD, and SPY (2004-2026). The plot shows periods of relative calm and high volatility, such as the 2008 Financial Crisis and the 2020 COVID-19 pandemic, affecting all assets.

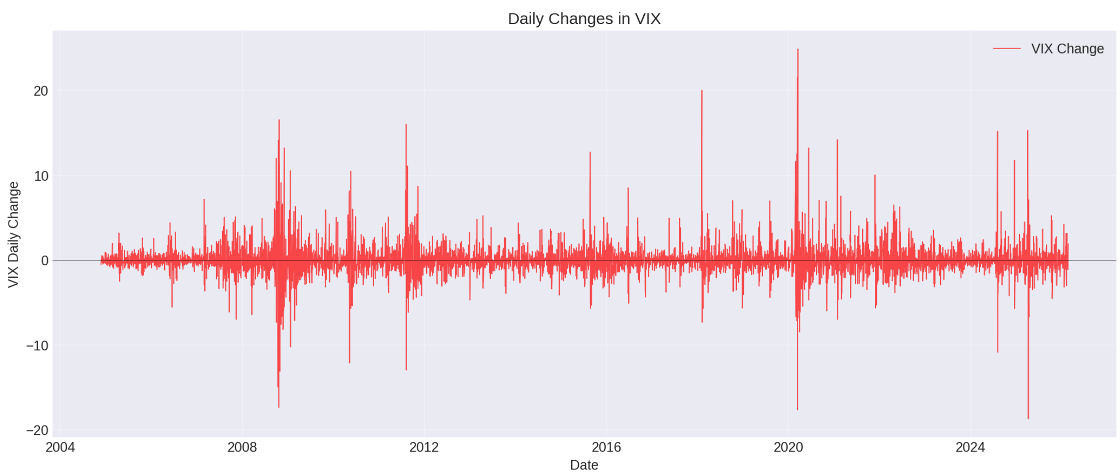


Figure 2: Daily Change in VIX ($\Delta$VIX) over the sample period. The series is highly volatile, with large positive spikes corresponding to market stress events.

**Interpretation:** The visual exploration confirms that VIX spikes are associated with major market dislocation events. The ETF returns also exhibit periods of varying volatility, suggesting that a regime-dependent model is a plausible approach.

# 3 Step 2 and 3: Modeling VIX Regimes and State Selection

## 3.1 Methodology

We explored two families of models to identify regimes from the $\Delta$VIX series:

1. **Discrete Markov Chain (MC):** $\Delta$VIX was discretized into 2 and 3 states using quantiles. A transition matrix and stationary distribution were estimated.
2. **Hidden Markov Model (HMM):** Gaussian HMMs with 2 and 3 states were fitted to the $\Delta$VIX series using the Expectation-Maximization (EM) algorithm. This approach treats the regimes as "hidden" states that generate the observed VIX changes.

To select the best model, we compared statistical fit criteria. Crucially, for economic validation, we analyzed the **next-day** returns of the ETFs conditional on the regime identified today. This forward-looking approach is essential for a tradable strategy.

## 3.2 Results and Model Selection

The Markov Chain models produced regimes with very low persistence (expected duration of 1.5-2 days), making them unsuitable for a practical trading strategy that requires a stable signal. The HMMs, in contrast, identified highly persistent states.

Table 1: Model Comparison for HMMs

| **Model** | **States** | **Log-Likelihood** | **AIC** | **BIC** |
|---|---|---|---|---|
| 2-State HMM | 2 | -9090.74 | 18193.47 | 18232.98 |
| **3-State HMM** | **3** | **-8745.51** | **17515.03** | **17594.04** |

The 3-State HMM was selected as the preferred model because:

- **Statistical Superiority:** It has the lowest AIC and BIC, indicating a better fit without unnecessary complexity.
- **Economic Interpretability:** The three states were clearly interpretable as Low, Medium, and High volatility regimes based on the mean and variance of $\Delta$VIX.
- **Regime Persistence:** The model implies economically meaningful expected state durations ranging from 8 to 16 days, which is suitable for a tactical rotation strategy.

Table 2: Characteristics of the 3-State HMM Regimes

| State | Mean $\Delta$VIX | Variance | Expected Duration |
|---|---|---|---|
| Low Vol (Calm) | -0.060 | 0.38 | 16.3 days |
| Medium Vol (Turbulent) | -0.014 | 2.99 | 11.1 days |
| High Vol (Crisis) | 0.628 | 35.46 | 8.0 days |

## 3.3 ETF Performance by State (Next-Day Returns)

Using the states identified by the 3-State HMM, we computed the mean next-day returns for each ETF. This forms the economic basis for our allocation rules.

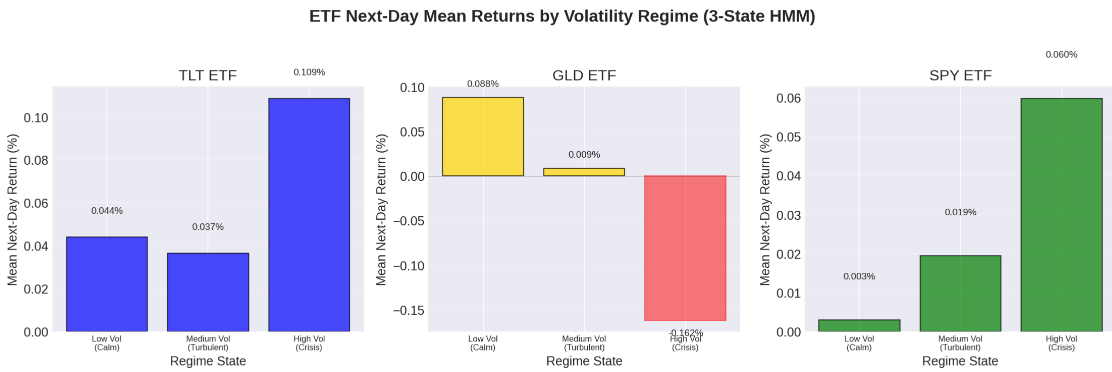


Figure 3: Mean Next-Day ETF Returns Conditional on Today's Volatility Regime.

- **Low Vol (Calm):** GLD is the best performer. This aligns with gold's role in a stable, low-opportunity-cost environment.
- **Medium Vol (Turbulent):** Performance is mixed. TLT leads, but SPY also shows positive returns, suggesting a need for diversification.
- **High Vol (Crisis):** TLT is the clear winner, acting as a classic safe-haven asset. GLD underperforms significantly. *Note: The positive return for SPY in this state is an anomaly relative to conventional theory and is discussed in Section 6.*

# 4 Step 4: Designing the Rotation Strategy

## 4.1 Allocation Rules

Based on the empirical results in Figure 3, we defined the following simple, rule-based allocation:

- **Low Vol (Calm):** 100% GLD (Best absolute performer).
- **Medium Vol (Turbulent):** 60% TLT / 40% SPY (Diversification to capture mixed signals).
- **High Vol (Crisis):** 100% TLT (Best performer and safe-haven asset).

To eliminate lookahead bias, a 1-day execution lag is implemented: the regime identified at the close of day $t$ determines the allocation for the trade executed at the open of day $t+1$, with returns captured from day $t+1$ to $t+2$.

## 4.2 Backtest Results

The strategy was backtested over the full sample period and compared to two benchmarks: an equal-weight (1/3 each) portfolio rebalanced monthly, and a simple buy-and-hold of SPY.

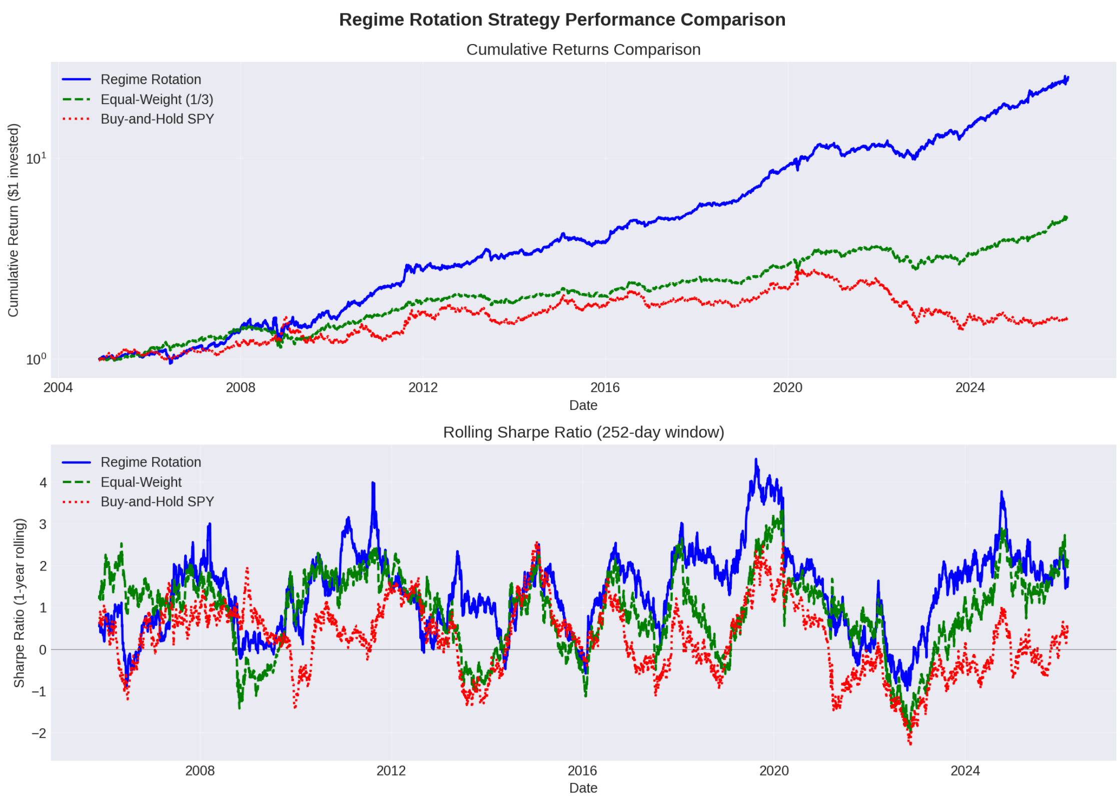


Figure 4: Cumulative Performance: Regime Rotation vs. Benchmarks (Log Scale).

Table 3: Performance Metrics Comparison

| Strategy | Cum. Return | Ann. Return | Ann. Vol | Sharpe Ratio | Max Drawdow |
|---|---|---|---|---|---|
| **Regime Rotation** | **2404%** | **16.40%** | 13.41% | **1.074** | **-23.03%** |
| Equal-Weight (1/3) | 412% | 8.00% | 9.71% | 0.618 | -23.99% |
| Buy-and-Hold SPY | 60% | 2.24% | 14.62% | 0.017 | -51.11% |

### 4.3 Analysis of Results

The regime rotation strategy dramatically outperformed both benchmarks.

- **Risk-Adjusted Returns:** With a Sharpe Ratio of 1.074, the strategy delivered far superior risk-adjusted returns compared to the S&P 500 (0.017).
- **Drawdown Protection:** The strategy successfully protected capital during major market downturns, with a maximum drawdown of -23%, less than half that of the S&P 500. This is a key benefit of rotating into safe-haven assets like TLT during crises.
- **Cumulative Wealth:** An initial investment of $1 in the rotation strategy would have grown to over $25, compared to just $1.60 for the S&P 500.

## 5 Step 5: Sensitivity Analysis and Discussion

### 5.1 Sensitivity Analysis

- **Number of States:** A 2-State HMM strategy also outperformed benchmarks but with a slightly higher Sharpe ratio (1.214) but also a higher max drawdown (-25.65%). The 3-state model provides more nuanced and economically intuitive allocations.
- **Sample Period:** The strategy performed well in both halves of the sample period, though performance was stronger in the second half (Sharpe 1.415) than the first (Sharpe 0.803). This suggests the strategy is robust but its characteristics may vary over time.

### 5.2 Limitations and Discussion

- **The SPY Anomaly:** Our analysis shows positive next-day returns for SPY in the "Crisis" state, which contradicts the typical "flight-from-risk" narrative. This could be due to the specific sample period including swift policy responses (e.g., post-2008, COVID-19) that led to sharp rebounds, which the "Crisis" state may inadvertently capture.

- **Model and Parameter Risk:** The HMM's parameters are estimated from historical data and are not guaranteed to be stable in the future. The model assumes Gaussian distributions, which may underestimate tail risk.

- **Generalizability:** The strategy is built on a specific sample period (2004-2026). Its performance in different market environments (e.g., a prolonged bear market not caused by a VIX spike) is unknown.

# 6 Conclusion and Practical Takeaways

## 6.1 Conclusion

This project successfully demonstrates the value of a regime-based allocation strategy using a Hidden Markov Model. By inferring volatility regimes from VIX changes, we constructed a dynamic portfolio that rotates among stocks (SPY), bonds (TLT), and gold (GLD). The backtest results show a substantial improvement in risk-adjusted returns and drawdown control compared to static benchmarks.

By applying a statistical model called a Hidden Markov Model (HMM) to changes in the VIX, we identified three distinct market regimes: **Calm**, **Turbulent**, and **Crisis**. Each regime exhibits a unique pattern of expected returns for the three assets. For example, Gold performs best in calm markets, while Treasuries act as a safe haven during crises.

We built an investment rule that allocates 100% to the best-performing asset in each regime (e.g., 100% TLT during a Crisis). A backtest comparing this "regime rotation" strategy against simply holding the S&P 500 or an equal-weight portfolio from 2004 to 2026 showed remarkable results. The rotation strategy delivered significantly higher returns with lower risk, most notably protecting capital during market downturns with a maximum loss of -23%, compared to a -51% loss for the S&P 500.

# References


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[3] Whaley, R. E. (2000). The investor fear gauge. *Journal of Portfolio Management*, 26(3), 12-17.